



Not like the others: Frontier scientists for inventive performance

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ABSTRACT

Linking scientific articles in PubMed and corporate biomedical U.S. patents, we study the role of inventors who are frontier scientists, identified as authors of recent articles in top-general biomedical journals. We find that inventions by these “frontier authors” receive more patent citations, are more likely to become technology hits, and have broad technology impact. They are also more likely to be internally further developed by the firm, hold greater private value, and feature broader claims—not only compared to inventions by non-author inventors but also to those by non-frontier authors, including “star” authors. This impact premium is especially strong in scaled-up young biopharmaceutical firms and for frontier authors internally employed at the patenting firm. To better understand the mechanism behind the impact premium of frontier-author patents, we analyze their boundary spanning role. We find that frontier-author patents are more likely to use and to be first users of frontier science. However, while frontier-author patents achieve peak impact when referencing frontier science, this advantage is comparable to other patents that reference frontier science. And as frontier-author patents also enjoy an impact premium on patents referencing other than frontier science, our results, thus, suggest that closeness to frontier science in only part of the story of the superior impact of inventions by frontier scientists.

1. Introduction

The economics of innovation literature has since long recognized the contribution of new scientific ideas to technological progress in industry (Nelson, 1962; Griliches, 1986; Jaffe, 1989; Mansfield, 1995; Cohen et al., 2002; Fleming and Sorenson, 2004), yet also made clear that the links between science and technology are neither obvious, direct nor immediate (among others, (Stephan, 1996; Narin et al., 1997; Ahmadpoor and Jones, 2017; Iaria et al., 2018)).

More recently, the literature has started to dig deeper into any timing advantages for industry science links (e.g., (Arora et al., 2023)), which science would matter most for technology development (e.g., (Poege et al., 2019; Veugelers and Wang, 2019)) and which individuals can effectively translate science into successful inventions. Specifically, inventor-authors, who cross the boundaries between scientific research and technology development, have been identified as important connectors for establishing industry science links (Gittelman and Kogut,

2003; Furukawa and Goto, 2006; Bonaccorsi and Thoma, 2007; Breschi and Catalini, 2010; Arts and Veugelers, 2020; Herrera, 2020).

Of particular interest is to understand the importance of “frontier science”—i.e., the newest science with big and broad impact, shifting the frontier of our knowledge—and which firms and inventors would be best at linking this frontier science into successful inventions.

In this paper, we examine whether corporate inventors who are at the scientific frontier—labelled as “frontier authors” from here on—are more impactful. We investigate what distinguishes corporate inventions by frontier authors from those of other inventors. We further assess for which firm, inventor and technology characteristics a frontier-author premium may be particularly pronounced.

To study the relation between frontier authors and impactful inventions we focus on the biomedical science-industry nexus. The biomedical industry is a relevant context given the strong connect between science and technology (e.g., (Gambardella, 1992; Cockburn and Henderson, 1998; Henderson et al., 1999; Murray, 2002; Li et al.,

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2017)). Our sample includes all U.S. biomedical patents assigned to firms between 1980 and 2009, along with all PubMed-indexed scientific articles published in Science Citation Index (SCI) journals. We cross-link authors and inventors relying on the disambiguations by Torvik (Torvik, 2018) and Li et al. (Li et al., 2014).

We operationalize *frontier science* as “recently” published in “top-general journals”, i.e., the most prestigious, general-interest-oriented journals in the biomedical discipline, and *frontier authors* as those inventors with “recent articles in top-general journals”.

We find that firm patents invented by frontier authors have higher technology impact (as measured by patent citations received) compared to similar patents without authors and, in particular, a much higher probability of being hit and broad patents. Importantly, frontier-author patents also have significantly higher impact than comparable patents by other authors lacking the frontier dimension—i.e., those who published in a top-general journal but not recently, those who published recently but not in a top-general journal, or those who did neither. These impact premia for frontier-author patents hold stronger for first authors and decrease substantially when using lower benchmarks for journal or article quality, when considering only field-specific top journals, and when including less recent publications. These findings are consistent with the idea that being active at the scientific frontier—by the recency and top-general nature of the science involved—matters particularly for fostering technological progress. We further find evidence that the superior performance of frontier-author inventions also holds, if not even more strongly—to generate higher private value for their firms. Additional analyses show that the impact premia for patents by frontier authors are strongest in young, scaled-up biopharmaceutical firms, and particularly when frontier authors are internally employed rather than when hired externally or when they are external collaborators.

Probing into possible mechanisms driving this pattern, we find that patents by frontier authors are not only more likely to cite scientific prior art in general—and do so more intensively—but are also significantly more likely to *cite frontier science*, compared to patents by non-authors and by authors who are not at the frontier. We further show that reliance on frontier science positively relates to more impactful inventions, a greater likelihood of technology hits, and higher private patent value. However, since this pattern holds for all patents referencing frontier science—not only those by frontier authors—and because frontier-author patents outperform even among those without frontier science references, their greater use of frontier science can only be part of the story of why frontier-scientists' inventions have superior impact. Finally, we find no evidence that technology impact premium of frontier-author patents depends on being first to cite frontier science. By contrast, priority does matter for their private value premium.

Taken together, our results extend the literature on the importance of corporate scientists and their impact on innovation in the biomedical industry by highlighting the unique contribution of *frontier author-inventors* compared to other corporate scientists. Corporate patents with frontier scientists as inventors are particularly effective in driving technological progress in industry, especially for young, scaled-up biotech firms. This effect is at least partially driven by these patents' ability to build on new scientific insights at the frontier of knowledge. Overall, our findings underscore the importance of unpacking the frontier nature of science for better understanding the role of inventor-authors and prior art references to science in predicting inventions with the highest technological impact and private value for firms.

2. A review of the literature and our hypotheses

2.1. Science for technology impact

Scientific advance and technological progress are long known to be closely coupled. Using forward citations as an indicator for technology impact, Fleming & Sorenson (Fleming and Sorenson, 2004), among others, show that patents that rely on scientific prior art have

significantly more impact on follow-on invention. Scientific inputs have also been found to yield inventions with superior private returns (Griliches, 1979; Mansfield, 1991; Krieger et al., 2024).

A growing empirical literature has started to investigate which type of science is most relevant for technology. Most of these analyses focus on assessing the relevance of high quality science, measured as being highly cited or published in top cited journals (Poege et al., 2019; Hicks et al., 2000).

2.2. Inventor-authors as knowledge brokers between science and technological progress in industry

Starting from Arrow (Arrow, 1962), innovation scholars have highlighted the existence of widely disconnected communities of scientists and inventors. Prior work has stressed the key role of knowledge brokers, i.e., individuals embedded simultaneously in the science and technology realms (Allen, 1977; Gruber et al., 2013).

A significant gap in the literature remains regarding our understanding of *which* inventors are best at brokering between science and technological progress in industry. Most of the literature in this field focuses on “academic scientists”, either in the context of academic entrepreneurship (Bercovitz and Feldman, 2008; Grimaldi et al., 2011; Bozeman et al., 2013) or in the context of firms collaborating with scientists (e.g., Cockburn and Henderson, 1998; Fabrizio, 2009)). As observed by several studies investigating the birth of the U.S. biotechnology industry (e.g., (Zucker and Darby, 1996; Zucker et al., 1998)), outcomes of firms collaborating with scientists are highly skewed toward “star” scientists (Zucker et al., 2002; Colen et al., 2022). Other studies examine “inventor-authors” on patents (Breschi and Catalini, 2010; Balconi et al., 2004). Gittelman & Kogut (Gittelman and Kogut, 2003) find that the number of follow-up citations of patents increases with the share of scientific authors among inventors on a patent.

Distinguishing between academic and corporate scientists is important due to the different knowledge production regimes in academia and industry. However, information on the exact link between the patenting firm and the listed inventors is often missing, explaining why less research has been conducted on corporate scientists (see Herrera (Herrera, 2020) for a survey). Furukawa & Goto (Furukawa and Goto, 2006), using a sample of five Japanese pharmaceutical companies, show that corporate scientists who publish numerous and frequently cited papers may not generate more patents themselves, but still create “brokerage” by positively influencing the patents of their co-authors. Arts & Veugelers (Arts and Veugelers, 2020), on a sample of surveyed Ph.D. graduates working in industry in Belgium, show that those who engage in academic boundary spanning—as proxied by scientific articles co-authored with academic scientists—produce more novel and impactful patents.

2.3. Our hypotheses: frontier authors and their use of frontier science for impactful inventions

Both scientific and technological progress are often depicted as advancing by a handful of big, new ideas or breakthroughs (Schumpeter, 1942; Kuhn, 1962; Dosi, 1982; Scherer and Harhoff, 2000; Wang et al., 2013). This immediately prompts the question of whether new, big ideas at the *frontier* of science are also more relevant for technological progress, and particularly for new technological breakthroughs. Research pushing the scientific frontier provides opportunities for new types of inventions, stemming from the translation of a new understanding of (new) fundamental principles into impactful technological solutions.

Frontier science is not only characterized by high quality but also by its recency and broad impact. The positive relationship between the recency of scientific involvement and technology outcomes has received some attention in prior studies (e.g., (Gruber et al., 2013; Fabrizio, 2009)). More recently, Poege et al. (Poege et al., 2019) linked high quality and recent science to technology impact by showing that the

influence of high-quality scientific articles is strongest when the time lag between the publication of the cited articles and the patent filing is shorter.

Identifying and building on frontier science to create high-impact inventions remains particularly challenging. Scientific insights at the knowledge frontier are especially difficult to absorb in technological invention due to the novelty and implicit nature of embodied ideas and methods. This calls for an even more explicit role of inventor-authors—knowledge brokers with roots in both communities—who can understand, analyze and assimilate disparate pieces of knowledge to leverage into technology development.

As we expect the scope for impact—and especially big impact—to come from the “frontier” of science, defined by the combination of excellence, breadth and recency of scientific insights, we single out “frontier authors” among inventor-authors on corporate patents. We expect these frontier scientists to have an edge not only over non-scientists but also over the typically examined “star”/excellent scientists. Indeed, as the frontier of knowledge is rapidly moving (e.g., (Mukherjee et al., 2017)), following the obsolescence of their insights, any impact advantage of excellent authors in technology development is likely to decay with time. Similarly, we compare frontier science to domain-specific scientific excellence to isolate the importance of the general and broad nature of frontier science impact.

We further explore the mechanisms underlying any inventive performance premium of frontier-author patents. We hypothesize that the specific advantage of patents with frontier authors stems from their boundary spanning role between science and technology development. More specifically, this advantage relates to their enhanced access to the rapidly evolving scientific frontier and their ability to use this knowledge as prior art for their inventions—most directly their own frontier knowledge, but not exclusively. We expect frontier authors to establish links to frontier science rapidly, often even before new scientific knowledge becomes public, thereby enjoying significant first-mover advantages. Arora et al. (Arora et al., 2023) find that citing scientific articles early correlates with higher private patent value, especially for articles with strong industrial applicability.

Our main hypotheses can be summarized as follows:

H1. : Firm inventions by frontier authors have greater technology impact than those by non-frontier authors or non-authors.

H2. : Firm inventions by frontier authors are more likely and faster to use frontier science as prior art than those by non-frontier authors or non-authors.

H3. : Firm inventions by frontier authors that use frontier science as prior art have greater technology impact than comparable inventions from non-frontier authors or non-authors.

We illustrate our research hypotheses with the case of the invention of high-density DNA microarrays. These microarrays have revolutionized genomics allowing to analyze the expression levels of thousands of genes simultaneously. Beyond ‘inherent’ fields like functional or environmental genomics, DNA microarrays are now widely used across many areas of medical research, including oncology, pharmacogenomics, or personalized medicine, where they help to predict biomarkers, assess drug responses, or characterize disease mechanisms. The original patent was filed in 1990 by Affymax N.V. (with the division that developed it later spun off as Affymetrix Inc.): U.S. Patent No. 5,143,854 became one of the most impactful patents in the history of the United States Patent and Trademark Office, receiving more than 3500 subsequent patent citations,¹ and Affymetrix—with its landmark product, the *GeneChip*®—the market leader in DNA microarrays.

The inventor team on this patent included neurobiologist Lubert Stryer. In the late 1980s, he achieved a major breakthrough by

discovering the fundamental metabiological mechanism by which retinal cells in the human eye perceive light. Two articles on this discovery were published in *Science* and *Nature* in 1988 (Meyer et al., 1988; Koch and Stryer, 1988). The profound understanding of principles acquired with this scientific breakthrough made Stryer rapidly realize that it should conversely be possible to harness the effect of light on macromolecules in cells to synthesize large amounts of compounds in a small number of chemical steps—the core idea behind DNA microarrays.

Stryer requested academic leave from Stanford University to collaborate with Affymax in pursuing his idea and developing a suitable technology. Soon after, the team discovered a process by which they succeeded in synthesizing 1024 peptides in only ten steps, as described in their influential article published in *Science* in 1991 (Fodor et al., 1991). This process was applied to the invention of miniature high-density arrays of biological compounds, for which Stryer and colleagues began filing a bundle of patents for Affymetrix in 1990—several of which became technology hits with broad impact.² Stryer’s seminal patent on high-density microarrays of biological compounds cited his own recent *Science* article on this new method, but he and his team also leveraged other frontier science for providing a highly successful new technological solution.

3. Research setting, data & methods

3.1. Data sources

We analyze the relationship between frontier authors, the use of frontier science as prior art, and technology impact for all biomedical utility patents assigned to firms and granted by the U.S. Patent and Trademark Office (USPTO) between 1980 and 2009. We link this sample to the USPTO disambiguated inventor database (Li et al., 2014). Our final sample comprises 237,345 patents invented by 159,994 unique inventors. We match inventors to authors and scientific non-patent references to articles in PubMed/SCI as detailed in the Supplementary material.

We further extract the titles and abstracts of all PubMed/SCI articles and assess their semantic overlap with the textual descriptions of all U.S. patents. Finally, we obtain article-to-article citation data from Microsoft Academic Graph (MAG; Sinha et al. (Sinha et al., 2015)) and use OpenAlex (Priem et al., 2022) to enrich PubMed author affiliation data. In the Supplementary material, we provide further details on data processing.

3.2. Measures

3.2.1. Patent impact

We utilize forward citations received to quantify the technology impact of patents in our sample. Forward citations are a commonly used and established indicator to assess the quality of patents (Trajtenberg, 1990; Harhoff et al., 1999). For each patent, we count the total number of citing patents within the USPTO over a ten-year moving window from filing of its worldwide first application (priority date).

To test whether the impact of frontier authors is even more pronounced for breakthroughs, we follow prior literature (e.g., (Ahmadpoor and Jones, 2017; Trajtenberg, 1990)) in identifying “hit patents” as those entering the top fifth percentile of the citation distribution within their primary patent class-year cohort. Similarly, for patents receiving forward citations, we additionally measure the breadth (or dispersion) of their impact across primary three-digit patent classes using the generality index from Trajtenberg et al. (Trajtenberg et al., 1997). We classify patents in our sample as having broad general impact if they fall

¹ U.S. Patent No. 5, 143, 854. Google Patents. Retrieved on 2024-04-08.

² Other patents related to this invention—in addition to U.S. Patent No. 5, 143,854—include U.S. Patent No. 5, 445,934, U.S. Patent No. 5,744,305, U.S. Patent No. 5,424,186, and U.S. Patent No. 5,405,783.

within the top quartile of this index.

We extend our analysis to other dimensions of patent value beyond technology impact, most notably the private value of patents for firms. To this end, we first measure the probability that a patent receives an “internal” forward citation—i.e., a prior art reference from a subsequent patent by the same firm. Next, we use data from Kogan et al. (Kogan et al., 2017) to obtain monetary estimates of patent value based on abnormal stock market returns following patent grant announcements.³ We also use patent renewal fee payments (Pakes, 1986; Schankerman and Pakes, 1986; Hall and Harhoff, 2012) to capture a lower bound of patent value, based on patent holders' revealed preferences. Additionally, we measure patent scope by the length of a patent's first independent claim, which serves as a proxy for both the strength of legal protection (Kuhn and Thompson, 2019; Marco et al., 2019) and the breadth of the inventive contribution (Akcigit and Ates, 2023).

3.2.2. Frontier authors

To identify authors at the frontier of science in our patent sample, we leverage the rich heterogeneity of inventors' author footprints in our data. We define frontier science based on three dimensions: (1) *recency*, (2) *top quality*, and (3) *broad impact*. We classify scientific articles to be *recent* if they were published no more than three years prior to the application of a focal patent.⁴ We consider scientific articles to be of *top quality* and *broad impact* if they are published in one of the eight most prestigious journals in the biomedical disciplines: **The New England Journal of Medicine, The Lancet, JAMA Journal of the American Medical Association, Nature, Science, Cell, Nature Medicine, and the BMJ British Medical Journal**. Formally, we selected these journals—which we denote as “top-general journals”—based on the following criteria: (1) ranking in the top fifth percentile of the average journal impact factor (JIF)⁵ within a biomedical field and in the ninetyeth percentile of average JIF across *all* fields in this category, and (2) covering a broad range of fields and topics, i.e., being general-interest. We surveyed a group of biomedical researchers to validate our selection of eight “top-general” biomedical journals. Articles in these journals account for a large share of groundbreaking scientific discoveries across fields and carry high supra-disciplinary relevance. Subsequently, we classify patents as “frontier-author patents” if at least one of their inventors authored a top-general journal article within the three years at the time of the first patent application.⁶ For example, we consider inventor Lubert Stryer a frontier author on the 1990 patent (with priority year 1989) covering the invention of DNA micro-arrays, owing to his 1988 articles in *Science* (Meyer et al., 1988) and *Nature* (Koch and Stryer, 1988). Accordingly, we classify the patent itself as a frontier-author patent. “Non-frontier-author patents” are those without frontier authors but with at least one inventor who has authored a non-frontier scientific article.

Within the group of non-general (top) journals, we further identify as *top-field* journals, those with a JIF score in the top 5 % of their respective biomedical field. This allows us to distinguish non-frontier science into

top-field science and non-top science. We further split each of these categories into recent and non-recent. Together, these distinctions help identify which “frontier” characteristics are most decisive for technology impact of science: its top-general nature and/or its recency. This also enables a contrast with the commonly used “star scientist” concept, typically defined by top journal publications that are not necessarily recent or general.

3.2.3. Frontier SNPRs

We trace the use of frontier science in inventions by examining the scientific prior art references made by patents in the sample.⁷ Although not a perfect measure, SNPRs proxy for a relation between the cited scientific idea as prior art input for the development of the citing invention (Tijssen, 2001; Roach and Cohen, 2013; Callaert et al., 2014), and are commonly used in the literature.

We rely on the patents-to-PubMed link to differentiate between types of SNPRs. Analogous to the definition of frontier authors, we classify patents as “frontier-SNPR patents” if at least one of their scientific references links to an article in a top-general journal published no more than three years before the patent priority-defining application. We further define binary indicators for subcategories based on timing (“first to cite”) and on inventors' proximity to referenced articles (“self-cites”).

As an alternative to relying on SNPRs to detect frontier science links in patents, we analyze semantic references to new scientific concepts in the summary description of inventions in the text body of patents. Similar to Iaria et al. (Iaria et al., 2018), we proxy “frontier-scientific terms” by non-common language terms that appear for the first time in titles and abstracts of recent scientific articles in top-general journals. For details on our text-based approach and the robustness of results, see the Supplementary material.

3.3. Sample and summary statistics

Almost 60 % of patents in our sample have at least one inventor with prior scientific involvement, confirming the strong connection between science and technology in the biomedical field. About 17 % of patents are invented by authors with prior top-general journal articles—hence would commonly be considered as “star scientists”—but only 7 % fall into the category of “frontier-author” patents, i.e., those with at least one inventor with *recent* top-general journal performance.⁸ The vast majority (75 %) of frontier authors do not switch their frontier-author status across patenting observations in our sample (Table SM3b–c). Most frontier-author patents (72 %) have a single frontier author, who is typically also the only “star” scientist on the team (cf. Table SM4a–c). These observations underscore the pivotal role of the frontier author in these firm inventions.

Table 1 reports descriptive statistics for our main hypotheses.⁹ Patents by frontier authors, are more likely to be outliers—i.e., to become hit patents—and have higher average monetary value than patents in the other two groups (all pairwise differences significant below the 1 % level). This holds even though they receive fewer forward citations on average than those by non-frontier authors, and especially fewer than patents by non-authors. Frontier-author patents are more likely and faster to cite scientific prior art, often referencing scientific articles that were not yet published at the time of the first patent filing. They are especially likely to cite frontier science. Additionally, they are much more likely to reuse new frontier science words in their invention

³ These data are available for only a subset of our sample (36 %), since Kogan et al. (Kogan et al., 2017) cover only patents from U.S.-listed public firms in the CRSP database.

⁴ The three-year cut-off is a relatively strict definition of recency, since the patent development process—the actual inventive search—usually precedes the filing for a patent by one to two years.

⁵ A five-year average (2013–2018) is used to compute JIF scores. While JIF scores may fluctuate over time, all defined top-general journals were consistently among the very top biomedical outlets across 1980–2009.

⁶ To identify the frontier nature of authorship, we rely on journal-based indicators of scientific performance rather than article-level citations. The advantage of using journal information is that it is unaffected by the future success of patents, which could otherwise influence the citations received by articles of successful inventor-authors. Nevertheless, we report in SM the robustness of our results using citation-based indicators of frontier status.

⁷ We exclude SNPRs added by patent examiners. However, this information is only available for patents granted after January 2001. Including examiner-added references does not affect our results.

⁸ In the Supplementary material, we briefly discuss the selection of authors into corporate invention and show that frontier authors tend to be more likely than other authors to become inventors on firm patents (see Table SM1).

⁹ For additional summary statistics, see Table SM2.

Table 1
Summary statistics.

Variables	All	Frontier-author patents	Non-frontier-author patents	No-author patents
Hit patent	0.07	0.08	0.07	0.06
\$-value (in mio USD)	20.52	24.80	22.38	16.73
Patent forward citations (10y)	17.28	14.15	16.39	18.93
Number of SNPRs	4.81	17.68	5.69	1.52
Frontier SNPR (top-general journal, last 3y)	0.09	0.40	0.10	0.03
Citation lag to SNPR (years)	2.08	−0.31	2.00	3.86
Citation lag to frontier SNPR (years)	0.37	−0.18	0.62	0.64
Reuse of frontier science word	0.05	0.14	0.05	0.03
Patent forward citations if frontier SNPR = 1	14.50	13.79	13.99	18.62
Patent forward citations if non-frontier SNPR = 1	15.92	13.61	14.73	20.08
Patent forward citations if no SNPR = 1	18.40	15.75	18.21	18.68
Share of patents	1.00	0.07	0.52	0.41

Notes: The table reports patent-level group means. The sample consists of all biomedical U.S. patents exclusively assigned to firms and granted between 1980 and 2009 ($N = 237,345$). For details on data sources and measures, see [Section 3](#).

descriptions. Patents with SNPRs receive fewer citations on average than other patents.

3.4. Estimation methods

To econometrically assess differences in technology impact between patents by frontier authors, non-frontier authors and non-authors (H1), we estimate the following generalized regression:

$$Y_i = \beta_1 \text{FrontierAuthor}_i + \sum_{j=1}^n \beta_n \text{NonFrontierAuthor}_{j(i)} + \mathbf{X}_i' \gamma + \delta_{c(i)t(i)} + \eta_{s(i)d(i)} + \epsilon_i \quad (1)$$

We outline our main specifications below, with further details provided in the Supplementary material. As a baseline specification, we estimate eq. (1) with Y_i capturing technology impact through counts of forward citations received by patent i . We further estimate eq. (1) using alternative outcome measures of patent value. Our main variable of interest is the dummy variable FrontierAuthor_i . We further include a set of n mutually exclusive category indicators capturing non-frontier authors type j on patents. To assess differences between frontier- and non-frontier-author patents, we compare β_1 with the β_n coefficients.

Our specifications control for a vector of patent-level covariates $\mathbf{X}_i$ that may affect both frontier-author status and patent impact. To account more specifically for variation within and across firm-level environments—particularly firms' innovation capacity—we include a series of dummies for assignee firms' patent-stock decile ranks. In robustness analyses, we also incorporate firm fixed effects and inventor- and author-level characteristics. Further details on control variables are provided in the Supplementary material.

To test predictions from our second research question (H2)—that frontier-author patents are more likely to use frontier science—we estimate eq. (1) with dependent variable Y_i as a binary indicator equal to 1 if a patent includes at least one frontier SNPR and 0 otherwise.

Finally, to test predictions from our third research question (H3), we estimate the following augmented regression equation:

Table 2
Frontier-author patents and technology impact.

Variables	PPML		
	(1)	(2)	(3)
	PatCit10	PatCit10	PatCit10
Author	0.138*** (0.008)		
Frontier author (top-general journal, last 3y)		0.260*** (0.018)	0.270*** (0.018)
Non-frontier author		0.129*** (0.009)	
Author, top-general journal, > 3y			0.217*** (0.014)
Author, top field-journal, last 3y			0.196*** (0.013)
Author, top field-journal, > 3y			0.132*** (0.015)
Author, no top-journal			0.074*** (0.010)
Patent class × year	Yes	Yes	Yes
Number of inventors	Yes	Yes	Yes
Firm patent-stock decile × Δ	Yes	Yes	Yes
Observations	237,114	237,114	237,114
Pseudo R ²	0.28	0.28	0.28

Notes: For details on data sources and measures, see [Section 3](#). Heteroscedasticity robust standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

$$Y_i = \sum_{k=0}^n \beta_n \text{FrontierAuthor}_i \text{SNPR}_{k(i)} + \sum_{k=0}^n \beta_n \text{NonFrontierAuthor}_i \text{SNPR}_{k(i)} + \sum_{k=1}^n \beta_n \text{NoAuthor}_i \text{SNPR}_{k(i)} + \mathbf{X}_i' \gamma + \delta_{c(i)t(i)} + \eta_{s(i)d(i)} + \epsilon_i, \quad (2)$$

where we decompose author-SNPR links into a set of n mutually exclusive binary indicators for patents by frontier authors, non-frontier authors, or no authors, each making the k^{th} type of SNPRs. The ks index patents with frontier SNPRs, non-frontier SNPRs and no SNPRs. The reference group consists of patents without inventor-authors and without SNPRs.

4. Econometric results

4.1. Frontier-author patents and technology impact

[Table 2](#) reports the estimation results for the relationship between technology impact of firms' patents and their frontier-author status (H1). Column 1 shows the baseline estimate for the impact differential between patents by *authors* (of any type) and non-author patents. Consistent with prior literature ([Gittelman and Kogut, 2003](#); [Bonaccorsi and Thoma, 2007](#)), the point estimate indicates that patents by inventors with prior scientific authorship receive, on average, 15 % more forward citations (significant below the 1 % level).

Our main result on differences in citation performance for patents by *frontier authors* (H1) is reported in Column 2. In contrast to the descriptive results in [Table 1](#), the point estimates from our econometric specification confirm H1, revealing an important technology impact premium for patents by frontier authors, who receive 30 % more follow-on citations than patents without authors (reference group). This premium is significantly larger than that for patents by non-frontier authors, which receive on average 14 % more citations than non-author patents.¹⁰ This result remains robust when alternatively defining

¹⁰ Both the point estimates and the difference in coefficients are statistically significant below the 1 % level (Wald test for equality of coefficients: $\chi^2(2) = 323.41$, $p < 0.001$).

frontier authors (ex-post) as authors of highly cited articles (Table SM3a and 5).

To more clearly isolate the *frontier* character of authorship—defined as the combination of top-general journal publication and recency, in contrast to “star scientists” solely defined by top-journal articles—in Column 3 we further split the category of non-frontier-author patents by quality and recency of inventor-authors' scientific footprints. The results highlight the importance of *both* recency and top-general quality in identifying the premium associated with frontier science inventor-authorship. With respect to recency, patents by frontier authors achieve substantially greater excess impact than those by authors who published in top-general journals more than three years earlier.¹¹ Regarding generality, the difference in impact premia between publishing in top-general journals and in top-field journals is significant and substantial, both for recent and non-recent articles.^{12,13,14} The starkest difference is observed relative to patents with non-top authors ($\chi^2(1) = 108.22, p < 0.001$).

The frontier-author patent premium remains highly robust when controlling for additional inventor characteristics, such as prior patenting experience, as well as for the number and characteristics of authors on the patent (see Table SM8). We further find that excess patent citations are significantly higher for patents whose inventors are listed as *first* authors on frontier science articles (see Table SM7). Notably, the co-inventors of frontier authors are typically non-frontier authors, underscoring the unique role frontier authors play within invention teams (Table SM4). Overall, these results confirm a significant technology premium for patents by recent authors in top-general journals: “frontier authors”. Among all authors—and even among star authors—frontier-author patents stand out.

In Table 3, we report estimates for impact dimensions beyond patent citations received. To test whether the presence of frontier authors on firm patents is particularly important for generating technology breakthroughs, we estimate the probability of a *hit patent*, defined as one in the top 5 % of citations within a class-year. The results imply a qualitatively analogous relationship, but with economically larger effects than for average impact: for patents by frontier authors, the excess likelihood of being a hit is about 64 % higher—roughly twice the increment estimated for patents with non-frontier authors (Column 1).¹⁵ Column 2 shows that the technology impact of frontier-author patents is in larger part *internalized* by assignee firms compared to patents by non-frontier authors and non-authors: the probability of a subsequent citation by the same firm is about twice as high for frontier-author patents compared to the baseline of non-author patents—an increment roughly double that observed for non-frontier-author patents.¹⁶ Similarly, Column 3 reports *private (monetary) values* as patent outcomes. Not only are our main results fully robust when using USD values from Kogan et al. (Kogan

Table 3

Frontier-author patents and alternative impact dimensions.

Variables	OLS		
	(1)	(2)	(3)
	Hit patent [0/1]	Internal PatCit10 [0/1]	Ln (\$-value)
Frontier author	0.035*** (0.002)	0.030*** (0.004)	0.459*** (0.025)
Non-frontier author	0.018*** (0.001)	0.018*** (0.002)	0.067*** (0.016)
Patent class $\times$ year	Yes	Yes	Yes
Number of inventors	Yes	Yes	Yes
Firm patent-stock decile $\times \Delta$	Yes	Yes	Yes
Observations	237,124	237,124	85,504
Adjusted R ²	0.03	0.06	0.22

Notes: For details on data sources and measures, see Section 3. Heteroscedasticity robust standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

et al., 2017), the estimated magnitude of the frontier-author premium is even significantly larger with respect to the private value for patent-holding firms.

In Table SM9, we further extend the set of indicators for patent value and impact to include “biggest hits” (i.e., patents in the top percentile of impact within their class-year), renewal fee payments, patent scope and the breadth of inventive contribution (as measured by claim length), and the breadth of technology impact (generality). Across these indicators, the performance premium for frontier-author patents is even more pronounced, while differences between non-frontier- and non-author patents remain consistently small or insignificant—particularly with respect to private values. Overall, these results confirm that frontier-author patents not only stand out by being more cited, but also and more specifically as breakthrough inventions with top and broad impact, which patent-holding firms are able to appropriate.

We further investigate the importance of technology and firm context for the impact premium of frontier-author patents. First, we provide split-sample estimates by technology fields (Hall et al., 2001) in Table SM10. The results show that the frontier-author premium holds across all major (medical) *fields*, but it is most outspoken in *biotechnology*—a field likely most closely related to frontier science. To further examine the influence of the *firm* environment, we re-estimate (1) with firm fixed effects. Table SM8 shows that the impact premium for frontier-author patents also holds within firms, but is smaller and not always significantly larger than that of other author patents. Exploring further the firm dimension, we differentiate firms in our sample into small and young versus large and established. Small firms are defined as those with fewer than 500 employees, as measured by the USPTO small entity indicator. Young firms are those whose first patent priority filing occurred no more than five years prior to that of the focal patent. In our sample, 17 % of patents are applied for by small firms and 26 % by young firms (Table SM11a).¹⁷ While accounting for only about one-fourth of patents in the sample, young firms generate 35 % of all frontier-author patents. This overrepresentation is most pronounced among scaled-up (i.e., large) young firms, which account for one in four frontier author patents. Our example of the development of blockbuster DNA microarrays by frontier author Lubert Stryer at Affymax/Affymetrix illustrates this match between frontier authors in young scaled-up

¹¹ Difference in coefficients significant below the 5 % level ($\chi^2(1) = 6.46, p = 0.011$).

¹² Differences between top-general and top-field journals are significant below the 1 % level, for both recent ($\chi^2(1) = 13.22, p < 0.001$) and non-recent articles ($\chi^2(1) = 20.62, p < 0.001$).

¹³ Most frontier authors (59 %) also publish articles in top-field journals, signaling that their frontier expertise often includes field-specific excellence. However, there is no significant difference in the technology impact of patents depending on whether the frontier author has complementary top-field publications, suggesting that this characteristic does not drive the frontier impact premium. Results from this analysis are reported in Table SM6.

¹⁴ Results remain robust under stricter definitions of top-field journals—e.g., when excluding the bottom 25 % of top-field journals or restricting the set to those that also fall within the top 10 % of top biomedical journals across fields, analogous to the selection criteria for the eight top-general journals.

¹⁵ The difference in coefficients is significant below the 1 % level (F(1,234,356) = 53.09, $p < 0.001$).

¹⁶ Difference in coefficients significant below the 1 % level (F(1,234,356) = 9.20, $p = 0.002$).

¹⁷ The majority (62 %) of young firms in our sample do not qualify as small entities—either because they are already independently large or because they are affiliated with larger entities.

Table 4
Heterogeneous effects by firm type and frontier-author affiliation.

Variables	PPML (1) PatCit10	Variables	PPML (2) PatCit10
Frontier author, large young firm	0.566*** (0.037)	Frontier author, internal employee	0.365*** (0.036)
Frontier author, large established firm	0.374*** (0.032)	Frontier author, other corporate	0.237*** (0.031)
Frontier author, small young firm	0.322*** (0.044)	Frontier author, other non-industry	0.222*** (0.025)
Frontier author, small established firm	0.370*** (0.070)	Non-frontier author	0.130*** (0.009)
Non-frontier author, large young firm	0.485*** (0.023)		
Non-frontier author, large established firm	0.243*** (0.022)		
Non-frontier author, small young firm	0.303*** (0.024)		
Non-frontier author, small established firm	0.146*** (0.026)		
No author, large young firm	0.320*** (0.023)		
No author, large established firm	0.141*** (0.021)		
No author, small young firm	0.113*** (0.024)		
Patent class $\times$ year	Yes	Patent class $\times$ year	Yes
Number of inventors	Yes	Number of inventors	Yes
Firm patent-stock decile $\times$ Δ	Yes	Firm patent-stock decile $\times$ Δ	Yes
Observations	237,114	Observations	236,981
Pseudo R ²	0.29	Pseudo R ²	0.28

biotech firm environments.¹⁸ Table 4 shows that the premium for frontier-author patents is largest in scaled-up young firms and significantly greater than in small young firms.^{19,20}

We further investigate how the relationship between frontier authors and firms shapes the impact of frontier-author patents. In principle, identifying inventor-authors' affiliations would allow a precise assessment of the match between firms and inventor-authors—distinguishing those employed by the patenting firm from those with external affiliations, such as university-based scientists. However, given the comprehensive scope of our sample, this information is not consistently available for all patents. For the subset of frontier authors, however, we were able to conduct a fully comprehensive, high-quality match between their article-based affiliations and the patent assignee firms.²¹ Among frontier-author patents in our sample, 30 % involve only internal

frontier authors, 31 % include at least one frontier author with external corporate ties, and 39 % involve only frontier authors from non-industry research environments. Column 2 of Table 4 splits frontier-author patents by affiliation type (“internal”, “other corporate”, “other non-industry”) and shows that all affiliation types enjoy a premium over non-frontier-author patents.²² The highest premium is observed for patents by frontier author-inventors exclusively affiliated with the firm (i.e., internal employees), showing an increment about two-thirds larger than that of other frontier-author patents.²³ As can be seen in Table SM11c, the superior performance of internal frontier authors holds across all firm types, but is, again, most pronounced in young large firms. In these firms, internal frontier-author patents have a significantly higher impact premium than in other firm types. Young large firms realize the largest excess for patents by internal frontier authors, closely followed by those with non-industry affiliations.

4.2. Frontier-author patents and their use of frontier SNPRs

In this section, we further zero in on potential mechanisms through which the impact premium of frontier-author patents materializes. Specifically, we examine whether this advantage is related to a different use of scientific knowledge as prior art—particularly frontier science—as a mechanism underlying their higher impact (H2–3).

Before examining whether differences in the use of frontier science as prior art can explain the impact premium of frontier-author patents, we first test whether frontier SNPRs are themselves a potential source of higher technology impact. Results reported in the Supplementary material (Tables SM12–14; 18) show that patents referencing recent top-general science articles receive on average about 35 % more follow-on patent citations than patents without any SNPRs—nearly twice the effect estimated for non-frontier science references. These results are fully robust when we measure frontier science links in patents by reuse of frontier science terms (Table SM 18). Priority—i.e., being first to cite a frontier science article—is not crucial for the technology impact of frontier science. The frontier-SNPRs premium also holds for private value. However, unlike the results for patent citations, priority does translate into higher monetary private value for frontier SNPRs (Table SM14, see also (Arora et al., 2023)).

4.2.1. Frontier-author patents and the likelihood of frontier SNPRs

Table 5 reports estimates assessing whether patents by frontier authors are more likely to include frontier SNPRs (H2). Column 1 shows that frontier-author patents are significantly more likely to reference scientific articles than patents by non-frontier authors or non-authors.²⁴ Column 2 reveals that, among patents citing science, those by frontier authors contain considerably more SNPRs on average.²⁵ Such disparity in the volume of SNPRs implies that, ceteris paribus, frontier-author patents would already stochastically be more likely to include frontier SNPRs. To evaluate whether they are also more selective toward frontier science in their scientific references, we estimate probabilities conditional on the number of SNPRs (Column 3). The results show that patents by frontier authors are considerably more likely to cite frontier science, even when holding constant the number of articles cited—more than twice as likely as patents without authors. This contrasts markedly with patents by non-frontier authors and, thus, supports access to frontier science as a potential mechanism underlying the superior performance of frontier-author patents.

¹⁸ Other prominent firms in this category with substantial numbers of frontier author patents in our sample include Human Genome Sciences (418), Chiron (84), and Genentech (48).

¹⁹ The difference in coefficients between frontier author and non-frontier author patents in large young firms is significant below the 5 % level ($\chi^2(1) = 5.63$, $p = 0.018$), while the difference between large young firms and small young firms among frontier author patents is significant below the 1 % level ($\chi^2(1) = 23.32$, $p < 0.000$).

²⁰ Table SM11b shows that the technology impact premium of frontier author patents holds across all firm types considered but is notably higher in large than in small firms and, in particular, higher in young firms than in older ones.

²¹ In the Supplementary material, we provide a detailed description of the matching procedure and validation.

²² All pairwise differences are significant below the 1 % level.

²³ The difference in coefficients with “frontier author, other corporate” is significant below the 1 % level ($\chi^2(1) = 7.58$, $p = 0.006$).

²⁴ The difference in coefficients is significant below the 1 % level (F (1,234,356) = 2245.29, $p < 0.001$).

²⁵ The difference in coefficients is significant below the 1 % level ($\chi^2(1) = 1187.65$, $p < 0.001$).

Table 5
Frontier-author patents and frontier-SNPR likelihood.

Variables	OLS	PPML		OLS	
		with SNPR subsample		with author subsample	
	(1)	(2)	(3)	(4)	(5)
SNPR [0/1]		# SNPRs	Frontier SNPR [0/1]	First Frontier SNPR [0/1]	No self-cits Frontier SNPR[0/1]
Frontier author	0.323*** (0.004)	0.868*** (0.020)	0.176*** (0.005)	0.064*** (0.008)	0.149*** (0.004)
Non-frontier author	0.163*** (0.002)	0.316*** (0.016)	0.004 (0.003)		
With frontier SNPR subsample				Yes	No
Author-level controls				Yes	Yes
DV subsample average	0.399	12.063	0.223	0.316	0.246
Patent class × year	Yes	Yes	Yes	Yes	Yes
Number of inventors	Yes	Yes	Yes	Yes	Yes
Number of SNPRs	No	No	Yes	Yes	Yes
Firm patent-stock decile × Δ	Yes	Yes	Yes	Yes	Yes
Observations	237,124	94,223	94,223	18,330	73,908
Adjusted / pseudo R ²	0.28	0.17	0.31	0.13	0.32

Notes: For details on data sources and measures, see [Section 3](#). Heteroscedasticity robust standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Among patents with authors and frontier SNPRs, we find that frontier-authors patents are significantly more likely to be the first to cite a given frontier science article (Column 4). A natural pathway through which frontier-author patents may more frequently draw on frontier science is self-citation ([Ahmadpoor and Jones, 2017](#); [Tijssen, 2001](#)). Indeed, more than one in three frontier-author patents with frontier SNPRs (36 %) involve self-citations. However, when we discard self-citations from the dependent variable and re-estimate the likelihood of making frontier SNPRs (in Column 5), patents by frontier authors remain significantly more likely to reference frontier science. This suggests that the elevated use of frontier science in frontier-author patents extends beyond self-citation and also includes the use of others' frontier science.²⁶

4.2.2. Differences in impact of frontier SNPRs for frontier-author patents

In this section, we examine whether the superior impact of frontier-author patents can be attributed to their more effective use of frontier science relative to other patents (H3). [Table 6](#) presents regression coefficients for technology impact, distinguishing frontier- and non-frontier-author patents according to whether they cite frontier science, non-frontier science, or make no scientific references. The point estimates in Column 1 show that frontier-author patents achieve their highest technology impact when citing frontier science as prior art—relative to their patents citing non-frontier science and, especially, those without SNPRs. While the difference for frontier author patents with patents lacking SNPRs is statistically significant, the difference

²⁶ In Table SM19, we show the robustness of these results when measuring (frontier) science links in patents by the reuse of new scientific terms rather than SNPRs.

Table 6
Frontier-author patents, frontier SNPRs and technology impact.

Variables	PPML		OLS	
	(1)	(2)	(3)	(4)
	PatCit10	Hit patent [0/1]	General ≥ pct75 [0/1]	Ln (\$-value)
Frontier author, frontier SNPR	0.401*** (0.034)	0.046*** (0.005)	0.069*** (0.007)	0.498*** (0.043)
Frontier author, non-frontier SNPR	0.345*** (0.035)	0.039*** (0.004)	0.028*** (0.006)	0.327*** (0.042)
Frontier author, no SNPR	0.217*** (0.031)	0.031*** (0.004)	0.026*** (0.007)	0.479*** (0.050)
Non-frontier author, frontier SNPR	0.364*** (0.027)	0.046*** (0.004)	0.033*** (0.005)	0.185*** (0.039)
Non-frontier author, non-frontier SNPR	0.250*** (0.017)	0.030*** (0.002)	0.008** (0.003)	0.099*** (0.029)
Non-frontier author, no SNPR	0.125*** (0.010)	0.016*** (0.001)	0.000 (0.002)	0.007 (0.020)
No author, frontier SNPR	0.372*** (0.041)	0.042*** (0.006)	0.045*** (0.009)	0.067 (0.059)
No author, non-frontier SNPR	0.211*** (0.018)	0.024*** (0.002)	0.026*** (0.004)	0.022 (0.031)
Patent class × year	Yes	Yes	Yes	Yes
Number of inventors	Yes	Yes	Yes	Yes
Number of SNPRs	Yes	Yes	Yes	Yes
Firm patent-stock decile × Δ	Yes	Yes	Yes	Yes
Observations	237,114	237,124	237,124	85,504
Pseudo / adjusted R ²	0.29	0.03	0.09	0.22

Notes: For details on data sources and measures, see [Section 3](#). Heteroscedasticity robust standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

between frontier author patents citing frontier science and citing non-frontier science is marginal and not statistically significant.²⁷ Moreover, the performance premium associated with frontier SNPRs is not exclusive to frontier-author patents: it also holds for patents by non-frontier authors and even for those without any author involvement. Although these other patents are significantly less likely to cite frontier science (cf. [Table 5](#)), when they do, they too benefit from a notable impact premium relative to their counterparts that cite other or no science.²⁸ Furthermore, while frontier-SNPR patents by frontier authors exhibit the highest average impact, this excess premium is not statistically different from that of frontier-SNPR patents by other inventors.²⁹ In other words, although frontier authors are more likely to produce

²⁷ Difference in coefficients between frontier author patents citing frontier science vs. no SNPRs significant below the 1 % level ($\chi^2(1) = 16.83$, $p < 0.001$), while vs. non-frontier SNPRs only below the 20 % level ($\chi^2(1) = 1.84$, $p = 0.175$).

²⁸ All differences in coefficients significant below the 1 % level: non-frontier-author patents citing frontier science vs. non-frontier SNPRs: $\chi^2(1) = 27.33$, $p < 0.001$; vs. no SNPRs: $\chi^2(1) = 80.97$, $p < 0.001$; no author patents citing frontier science vs. non-frontier SNPRs: $\chi^2(1) = 16.35$, $p < 0.001$.

²⁹ Both the differences between frontier author patents with frontier SNPRs and non-frontier-author patents with frontier SNPRs ($\chi^2(1) = 1.44$, $p = 0.229$) as well as non-author patents with frontier SNPRs ($\chi^2(1) = 0.41$, $p = 0.521$) are not significant below the 10 % level

high impact frontier-SNPR patents (cf. section 4.2.1), the premium on such patents is not significantly greater than the frontier science premium observed for other inventors. In addition, since frontier-author patents also show an impact premium on non-SNPR patents,³⁰ access to frontier science can only be part of the mechanism explaining their superior performance.³¹

Similar patterns emerge when examining the likelihood of producing hit patents (Column 2). Noteworthy are the results on citation generality (Column 3): frontier-SNPR patents by frontier authors show a significantly broader impact than those from any other inventor-author science-link combination.³² When examining private patent value (Column 4), we find that the frontier-SNPR premium is larger for frontier-author patents than for those by non-frontier authors or non-authors. These differences are more pronounced and statistically significant, yet not significantly larger than those for frontier-author patents without SNPRs.³³ Superior use of frontier science (H3) thus appears more strongly supported as a mechanism for the private-value premium of frontier-author patents than for their higher technology impact. Nevertheless, even for private value, access to frontier science explains only part of the premium, as also frontier-author patents without frontier SNPRs—and even those without any SNPRs—still exhibit a significant private-value advantage compared to other patents.

Although frontier-author patents are more likely to be first to cite a frontier science article (cf. Table 5, Column 4), further analysis (Table SM16) suggest that their impact advantage from using frontier science does not primarily stem from establishing *priority* to frontier SNPRs. For private value, however, frontier-author patents with priority citations to frontier science hold a premium over those by non-frontier authors. Yet they exhibit an even larger private-value premium on patents citing frontier science *without* priority—suggesting that there is more to the story of the private-value premium for frontier-author patents than could be explained solely by privileged early access.

Further analysis of self-citations in frontier-author patents linked to frontier science shows that those citing their own frontier articles attain a significantly higher technology impact premium than those citing others' frontier science (Table SM17). In fact, a significant technology impact premium of frontier- over non-frontier-author patents (citing frontier science) emerges only for patents with self-cited frontier articles. For these patents, the prediction of H3 is confirmed—access to one's own frontier science appears to be a key lever for higher-impact inventions. In contrast, for non-self-cited frontier science links, there is no significant technology premium for frontier-author patents. However, in terms of private value, frontier-author patents also show a significant premium even when citing non-self-authored frontier science.

5. Conclusions

In this paper, we contribute to the literature on industry-science links by identifying and illustrating the advantage for firms in the biomedical

industry of having scientists active at the scientific frontier as inventors on their patents. Unlike the commonly studied highly cited scientists, *frontier scientists* produce science that is not only excellent but also recent and broadly impactful. We show that corporate inventions developed by frontier scientists achieve significantly higher and broader technology impact, exceeding that of inventions by non-authors as well as by top authors further away from the frontier, whether in time or general quality. This impact premium is particularly pronounced in scaled-up young firms and when frontier scientists are closely connected as internal employees of the patenting firm.

We further show that the citation premium of frontier-author patents is partially driven by their ability to identify and build on new scientific insights at the frontier of knowledge, as reflected in their greater inclination and intensity to use science as prior art—particularly the use of frontier science, including but not limited to their own. Although frontier authors possess a first mover advantage in accessing frontier science, we find no evidence that this advantage translates into higher technology impact. Moreover, since the impact premium of frontier-author patents also holds for inventions without links to science, our results suggest that their superior performance is a bigger story than access to frontier science alone.

Our main results are robust to several alternative approaches, including the use of text-mining techniques rather than scientific references (SNPRs) to identify reliance on new scientific ideas in corporate patents. When examining the private value of patents rather than citation impact, the superior performance of frontier-author patents and frontier-SNPR patents is not only confirmed but appears even more pronounced. Frontier-author patents are also significantly more likely to be used internally by firms as prior art for subsequent technology developments. For private value, access to frontier science—including priority access—provides a significantly higher premium for inventions by frontier authors, even when citing non-self-authored frontier science. Yet, much like the technology impact premium, the superior private value of frontier-author inventions cannot be explained solely by priority access to frontier science.

Our findings primarily contribute to the literature on the returns from science for firms in technology development (Gittelman and Kogut, 2003; Bonaccorsi and Thoma, 2007; Gruber et al., 2013). We emphasize the importance of the *frontier nature* of both the inventor-author link and the scientific prior art used—distinguishing it from other types of science, including top-field science. We show that frontier inventor-authors act as distinctive “bridge-builders” between science and technology (Breschi and Catalini, 2010). The superior inventive performance of frontier-author patents, compared to those of other inventor-authors, is at least partly driven by their ability to use frontier science, including—but not limited to—their own frontier science.

Our findings also contribute to the literature on firm participation in scientific research (Gambardella, 1992; Rosenberg, 1990; Arora et al., 2021). We highlight the heterogeneity of firm environments that are most conducive to realizing a frontier-author patent impact premium. Young, scaled-up firms—typically in biotechnology—appear best positioned to benefit from frontier inventor-authors, with the premium being significantly larger when the frontier author is an internal employee of the patenting firm.

Our findings also speak to recent work on channels of *early* access for firms to new scientific findings (Arora et al., 2023; Baruffaldi and Poege, 2025). Early access to frontier science pays off, but only in terms of private value, not citation impact. While frontier-author patents do benefit from early access to frontier science—including beyond their own—they also exhibit an impact premium without first access, suggesting that priority to frontier science is only part of the explanation for the superior performance of frontier-author inventions.

Overall, our findings strongly support the view that involving inventors active at the frontier of science enables firms to generate commercial inventions with superior technology and higher private value. The high likelihood of frontier-author patents to lead to big and broad

³⁰ Both pairwise differences with non-frontier author, non-frontier SNPR ($\chi^2(1) = 9.19, p = 0.002$) and no author, non-frontier SNPR ($\chi^2(1) = 15.94, p < 0.001$) are significant below the 1 % level.

³¹ Results are confirmed when using semantic detection of frontier science links through reuses of new scientific words in patent text for the same set of outcomes (cf. Table SM20).

³² Differences in coefficients are significant at the 1 % level when comparing frontier-author patent with frontier SNPRs to those with non-frontier SNPRs ($\chi^2(1) = 28.34, p < 0.001$) and to non-frontier-author patents with frontier SNPRs ($\chi^2(1) = 27.95, p < 0.001$) and at the 5 % level when compared to non-author patents with frontier SNPRs ($\chi^2(1) = 5.44, p = 0.020$).

³³ The difference between the first two coefficients in Column 4 of Table 6 is significant below the 1 % level ($\chi^2(1) = 16.03, p < 0.001$). However, the difference between frontier-author patents with frontier SNPRs and frontier-author patents without SNPRs is not significant below the 10 % level ($\chi^2(1) = 0.09, p = 0.759$).

technology breakthroughs on which many others can build also makes these patents of high social and policy relevance.

Our findings call for further research to better understand the sources of the performance premium for frontier science-based inventions—both within and beyond the biomedical industry. This includes a deeper investigation into who these frontier scientist-inventors are and what intrinsic advantages they may hold as generators of successful inventions. Beyond their access to and use of different types of science, other mechanisms need to be explored to explain their superior performance. Further research is also needed to examine which firm and technology environments are most conducive to realizing a frontier-author premium. Future work should clarify the matching between scientists and firms—who to employ and/or collaborate with—and investigate the selection of frontier authors into teams of inventors.

CRedit authorship contribution statement

Thomas Schaper: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sam Arts:** Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Reinhilde Veugelers:** Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Given her new role as editor of Research Policy, during the process of reviewing the article, Reinhilde Veugelers had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was with another journal editor. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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Data availability

Data will be made available on request.

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